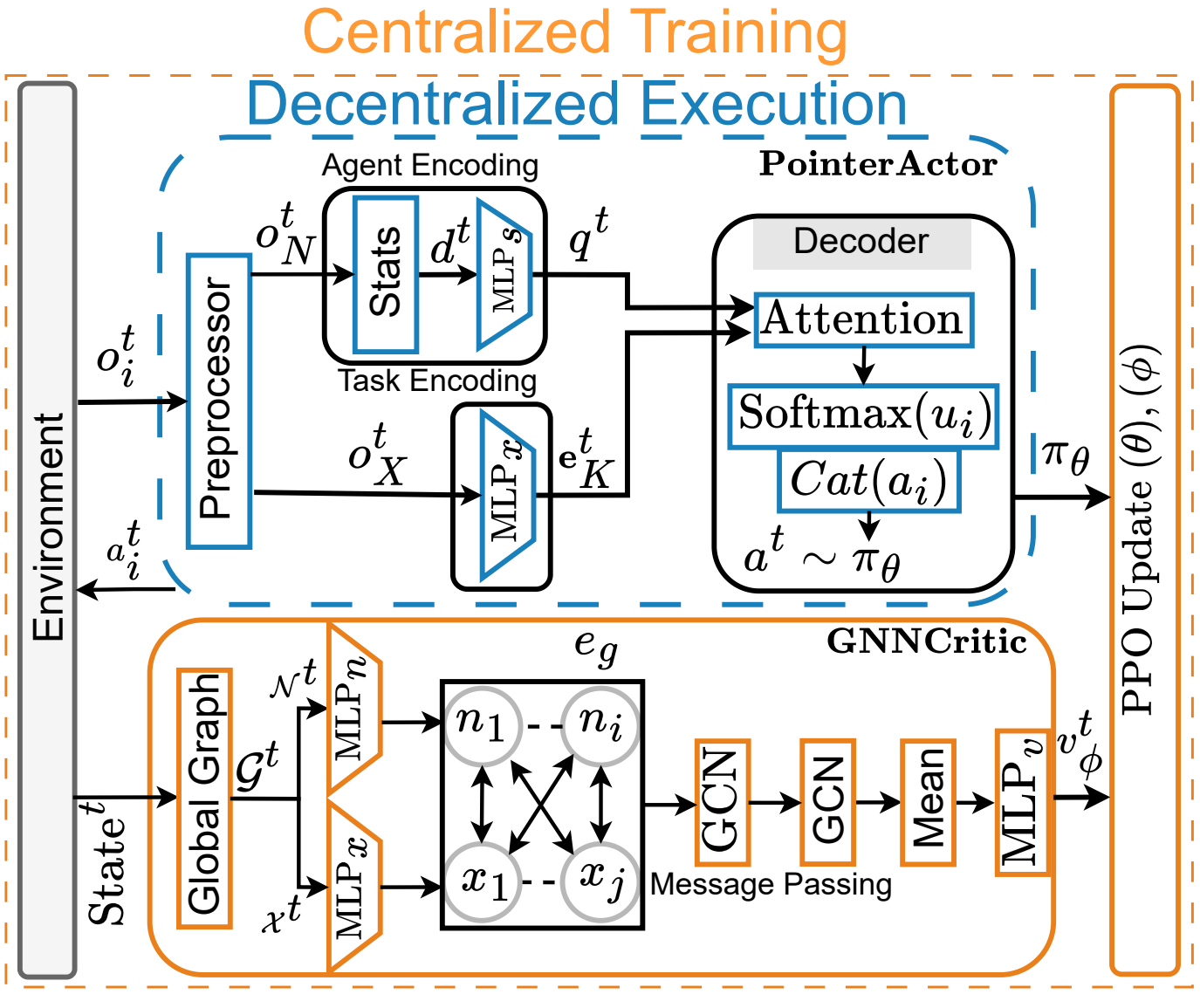

Centralized Training
Decentralized Execution
Agent Encoding
PointerActor
Environment
Preprocessor
Stats
MLP$_s$
Task Encoding
MLP$_x$
Decoder
Attention
Softmax(u_i)
$Cat(a_i)$
$a^t \sim \pi_\theta$
π_θ
PPO Update (θ), (ϕ)
GNNCritic
Global Graph
MLP$_n$
MLP$_x$
e_g
n_1 n_i
x_1 x_j
Message Passing
GCN
GCN
Mean
MLP$_v$
State^t
o_i^t
a_i^t
o_N^t
d^t
q^t
o_X^t
$\mathbf{e}_K^t$
$\mathcal{N}^t$
$\mathcal{G}^t$
x^t
v_ϕ^t